



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Subject : Database (SECD 2523)

Section : 03

Task : Phase 1 - Project Proposal and Planning

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1.0 Introduction

This report presents the proposal for developing a “Parcel Service System” for Cengal Parcel Point (CPP) at Universiti Teknologi Malaysia (UTM). Cengal Parcel Point plays an essential role in handling, storing, and distributing parcels for students and staff within the university. However, the existing process is largely manual and relies heavily on human effort, resulting in inefficiencies such as delayed notifications, misplaced parcels, and storage congestion during peak delivery seasons.

To address these challenges, this project proposes an automated, database-driven Parcel Service System designed to enhance efficiency, accuracy, and user satisfaction. The system incorporates a centralized MySQL database to manage parcel data systematically, ensuring reliable data storage, retrieval, and updating. Through automation, key processes such as parcel logging, notification delivery, tracking, and reporting are handled digitally, reducing staff workload and minimizing human error.

Additionally, the proposed system integrates QR code verification for secure parcel collection and a smart locker system to enable 24/7 self-service pickup, improving accessibility and convenience for users. Students and staff can also view real-time parcel statuses through a dedicated online portal, ensuring transparency and smoother operations.

In summary, this project aims to modernize the parcel management process at CPP by leveraging database technology and automation tools. The proposed system will streamline parcel handling, improve communication efficiency, and provide a more secure and user-friendly experience for both staff and students at Universiti Teknologi Malaysia.

2.0 Background Study

In this era where online shopping is extremely convenient and cost-effective, the demand for parcel delivery services are getting higher recently. Within the University Teknologi Malaysia (UTM), parcels ordered by students and staff will gather at parcel collection points like Cengal Parcel Point, which is mostly Shopee online shopping platform's parcel goes. This has become an essential and significant place for managing the growing volume of deliveries and ease for delivery partners with no need to deliver in large university areas.

Despite its benefits and importances, the current system of CPP is highly dependent on human resources and proceeds manually such as manual notifications. This has increased the workload of the staff and lowered the efficiency of the system. As the parcel volume increases, the system will face heavy workload, often resulting in:

- High risk of human error and misplaced parcels
- Increased workload and stress on CPP staff
- Students don't have time to collect during study time
- Inconsistent notification methods
- Storage congestion due to uncollected parcels

To better understand these problems and challenges faced, we have conducted an interview with the staff at CPP.



Figure 2.1: Interview session with staff at CPP

Key questions included:

1. How many parcels do you usually receive in one day? When is the busiest time?

-Normally about 100 parcels per day. Exam seasons and promotional sales like 11.11 and 12.12 will be the busiest period.

2. How do you currently notify students or staff when their parcel has arrived? (e.g., WhatsApp, paper slips, or phone calls?)

-Whatsapp notification is sent once a parcel is ready to pick up but sometimes will send wrongly the message causing the parcel didn't be picked up.

3. Would a 24/7 self-pickup machine or automated message system help? Why or why not?

- Yes, it will be really helpful as many students or staff are not available at the daily working period.

These insights show the need for a more automated and systematic Parcel Service System that integrates modern technology to boost up the efficiency by optimizing the operations, reducing human error, and improving customer satisfactions.

3.0 Problem Statement

Cengal Parcel Point (CPP) at Universiti Teknologi Malaysia is a vital and significant service center to handle the growing volume of parcels ordered by staff and students. After research and understanding the problems by interview session, the current parcel management system is found is not enough to cover daily workload as it most depends on human resources and is largely manual. The outdated approach led to several recurring problems, especially during peak volume delivery periods.

Key challenges:

- High workload on staff which need them to manage all parcel logging and notify recipients to collect their parcel once ready.
- Limited collection hours which restricted access and efficiency, reducing convenience for recipients at the same time.
- Inefficient parcel tracking and processing due to lack of digital records that will cause parcel missing issues
- Overcrowded storage areas caused by slow parcel turnover and delayed pickups process.

These problems have affected the operational efficiency and impact customer satisfaction and trust toward the parcel management system. CPP staff agreed and expressed a strong need for a more automated and reliable system that helps them to manage parcel collection processes by not only depending on human works.

Therefore, a digital and modern Parcel Service System including automated notifications, real time tracking, and 24/7 self service lockers to pick up will be a significant useful and beneficial system. An efficient system would help to optimize storage, improve communication between users and staff, enhance security and largely reduce staff's burden. Ultimately, the system can lead to a better experience for both users and administrations of the service.

4.0 Proposed Solutions

The proposed Parcel Service System is designed to enhance the operational efficiency and service quality of Cengal Parcel Point (CPP) at Universiti Teknologi Malaysia by replacing the current manual system with a fully automated and digitalized solution. The system focuses on improving efficiency, security, and accessibility for both CPP staff and parcel recipients. At present, most parcel management activities rely on manual entry and WhatsApp notifications, which often lead to misplaced parcels, delayed communication, and heavy staff workload. The proposed solution addresses these issues by automating key processes such as parcel logging, notification delivery, parcel tracking, and data reporting.

The system introduces five major functions. First, the Parcel Logging System allows CPP staff to record each parcel digitally into a centralized database. Each parcel entry includes details such as customer name, UTM ID, phone number, parcel size, and parcel ID. Once data is recorded, the system automatically generates a unique QR code for each corresponding parcel, which will later be used by the recipient for verification and collection. This function ensures data accuracy, traceability, and reduces the risk of human error.

Second, the Automated Notification Feature improves communication between CPP and students. After a parcel is logged, the system automatically sends notifications via email, SMS, or in-app alerts. Each message contains parcel details and a unique QR code that serves as a secure identifier during the pickup process. This automation eliminates the need for staff to manually send messages, thereby saving time and ensuring immediate delivery of notifications.

Third, the Student Parcel Status Portal provides students with a convenient and transparent way to track their parcels. By logging in with their UTM ID, users can view real-time parcel status such as Arrived, Ready for Collection, or Collected. The portal also displays penalty details for overdue collections. This feature minimizes walk-in inquiries, reduces congestion at CPP, and enhances user satisfaction by offering easy access to parcel information anytime and anywhere.

Fourth, the Smart Locker Integration introduces a self-service collection method that enables 24/7 accessibility. Based on parcel size and recipient address, the system automatically

assigns parcels either to the CPP counter or to a nearby smart locker based on the size of the parcel. Students can collect their parcels by scanning the QR code received through their notification. Once the QR code is scanned, the system automatically updates the parcel status to “Collected.” This integration promotes flexibility and convenience, especially for students living far from CPP, and helps to reduce overcrowding at the main parcel collection point.

Lastly, the Admin Dashboard and Storage Monitoring feature gives CPP staff a powerful tool to manage operations more effectively. The dashboard provides real-time insights into parcel flow, locker capacity, overdue pickups, and feedback. It also supports data visualization and report generation on a daily, weekly, or monthly basis, enabling better decision-making and workload management. Collectively, these components ensure that the Parcel Service System delivers a streamlined, secure, and scalable solution for parcel management at Cengal Parcel Point.

4.1 Technical Feasibility Study

The Parcel Service System is technically feasible as it relies on proven and readily available technologies. The system will be developed as a web-based application with database integration, QR code generation, and smart locker connectivity.

A Relational Database Management System (RDBMS) such as MySQL will be used to manage parcel and user data efficiently. The system architecture will follow a client-server model, allowing staff, students, and administrators to access the platform from multiple devices simultaneously.

For parcel identification and collection, the system will integrate QR code scanning technology, ensuring security and authenticity in every transaction. The smart locker units will communicate with the main system through secure APIs, enabling automatic parcel status updates once a package is retrieved.

Additionally, the admin dashboard provides real-time monitoring of storage capacity, overdue parcels, and locker usage. The use of HTTPS protocol ensures secure data transfer between all components.

Overall, the system's use of familiar technologies, cloud-based database hosting, and QR integration ensures that it is technically practical, secure, and expandable within the current infrastructure of Universiti Teknologi Malaysia.

4.2 Operational Feasibility Study

From an operational standpoint, the proposed system is highly feasible and will significantly enhance CPP's daily workflow. The current manual parcel management process requires staff to record details, send notifications manually, and handle large volumes of parcels physically. This results in high workload, delays, and human errors.

With the introduction of the Parcel Service System, most of these repetitive processes will be automated. The system will handle parcel registration, notification delivery, and parcel tracking automatically, allowing CPP staff to focus on exception handling and customer service instead of manual record keeping.

Students will benefit from real-time parcel status tracking and 24/7 access via smart lockers. This flexibility improves convenience, reduces overcrowding at CPP, and minimizes collection delays.

Furthermore, the system is designed with simple and intuitive user interfaces, ensuring ease of use for both students and staff with minimal training required. The digital workflow will also improve accountability, as every parcel transaction is recorded and traceable in the system's database.

In summary, the operational implementation of this system is both realistic and beneficial, as it aligns with CPP's existing work procedures while optimizing efficiency and service quality.

4.3 Economic Feasibility Study

Estimated Cost	Amount (RM)	Frequency
Hardware (locker banks)	50000	one-off
Software	10000	one-off
Consultant	5000	one-off
Training	5000	one-off
Annual maintenance/support	10000	per year
IS Support	2500	per year

Table 4.3.1: Estimated Cost

Estimated Benefits	Amount (RM)
Storage Fee (Within Period)	21000
Penalty Revenue (Overdue)	9000
Misplaced-handling & Time saving	20000

Table 4.3.2: Estimated Benefit

Assumption	Value
Discount rate	5%
Sensitivity factor (cost)	1.1
Sensitivity factor (benefits)	0.9
Annual change in production costs	5%
Annual change in benefits	7%

Table 4.3.3: Assumption for 5 years project duration

Cost	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Development Costs						
Hardware (locker banks)	55,000					
Software	11,000					
Consultant	5,500					
Training	5,500					
Total	77,000					
Production Costs						
Annual maintenance/support		11,000	11,550	12,128	12,734	13,371
IS Support		2,750	2,888	3,032	3,183	3,343
Annual Production Costs		13,750	14,438	15,159	15,917	16,713
(Present Value)		13,095	13,095	13,095	13,095	13,095
Accumulated Costs		90,095	103,190	116,286	129,381	142,476
Benefits						
Storage Fee (Within Period)		18,900	20,223	21,639	23,153	24,774
Reduced Penalty Revenue		8,100	8,667	9,274	9,923	10,617
Misplaced-handling & Time Saving		18,000	19,260	20,608	22,051	23,594
Total		45,000	48,150	51,521	55,127	58,986
(Present Value)		42,857	43,673	44,505	45,353	46,217
Accumulated Benefits (Present Value)		42,857	86,530	131,036	176,389	222,606
Gain or Loss		- 47,238	- 16,660	14,750	47,008	80,130
Profitability Index	1.04					

Table 4.3.4: Cost-Benefit-Analysis(CBA) Calculation

The table above shows the result of CBA for the CPP System. The project requires an initial development cost of RM77000 and the annual production costs for maintenance and support over a five year duration. The system is expected to generate benefits from storage fee, reduced penalty revenue and misplaced-handling & time saving with an accumulated present value of approximately RM222606. The analysis shows that in the beginning the project will have initial losses in Year 1 (RM47238) and Year 2 (RM16660), then the project begins to achieve positive number in Year 3 and continues to grow steadily, it will reaching a total gain of RM80130 in Year 5. The calculated profitability index of 1.04 stated that the project's benefits outweigh its total cost. To sum up, the result suggests that the CPP System is financially viable.

5.0 Objectives

The implementation of the digital Parcel Service System is to resolve some key issues that are faced by CPP. The specific goals of the system are:

I) Automate Parcel Management Tasks

The implementation of the smart lock and introducing a digital platform for parcel logging, notifications, and tracking can reduce staff workload wisely.

II) Enhance Notification Accuracy

The system can ensure an accurate delivery of parcel arrival messages timely through automated communication channels to the student.

III) Improve Parcel Security and Tracking

The implementation of QR code in verification and system logs stage can reduce parcel misplacement and increase accountability.

IV) Increase Accessibility for Users

24/7 smart lockers allow flexible collection hours via self-service methods, especially for students staying in remote colleges and the student having transportation problems.

VI) Optimize Storage Utilization

This system also enables better control over storage space and prevents congestion by speeding up pickup through alerts and reminders. The penalty will be charged for the overdue parcel to avoid the locker being occupied for a long time.

VII) Boost User Experience and Satisfaction

This system offers a user-friendly, transparent, and efficient process from parcel arrival to collection, improving overall satisfaction for both staff and students.

VIII) Integrate a Centralized Database

Implement a MySQL database to store and manage parcel, user, and transaction data securely, enabling easy retrieval, updates, and report generation.

6.0 Scope of the Project

The scope of this project focuses on designing and implementing a web-based Parcel Service System with full database integration to digitalize parcel management at Cengal Parcel Point (CPP), Universiti Teknologi Malaysia (UTM). The system is designed to improve the overall efficiency, security, and accessibility of parcel handling operations.

System Coverage:

- Enable automated parcel logging, notification, and collection tracking.
- Provide real-time parcel status updates through an online portal.
- Generate QR codes for secure parcel collection and verification.
- Support automated reports for daily, weekly, and monthly parcel management.
- Integrate with smart lockers to allow 24/7 self-service parcel pickup.
- Maintain a centralized MySQL database to manage users, parcels, and transactions.
- Ensure data security and controlled access through authentication and role-based privileges.

User Roles:

- **CPP Staff :** Manage parcels, update parcel statuses, monitor storage and locker usage, and generate reports.
- **Customer (Recipients include UTM Staff and Student):** Check parcel status, receive notifications, make payments, and collect parcels using QR codes.
- **Administrator:** Manages the entire system like user accounts, data maintenance, reports, and configurations.

Automation Focus:

- Replace manual records with a MySQL database for efficient data management.
- Support automated notification delivery via email or portal alerts.
- Provide data integrity through controlled access, validation, and secure backups.

Pickup Optimization:

- Introduce options for 24/7 self-service lockers to ease access and reduce congestion.

Expected Outcome:

- Reduce manual workload and human error in parcel handling.
- Improve storage utilization and prevent congestion.
- Enhance communication and transparency between CPP and parcel recipients.
- Provide convenient, flexible, and secure parcel collection experiences for all users.

Out of Scope (What the system will not cover):

- Integration with external courier services' systems (e.g., Shopee, J&T API integration).
- Mobile app development (only web portal included in this phase).
- Parcel delivery within the campus or to hostel rooms.
- Real-time GPS tracking of parcels during delivery.
- Allocate the parcels based on the size
- The system is limited to CPP only and does not extend to other parcel points within UTM.

Assumptions and Constraints:

- The system will be accessible through desktop or mobile web browsers only.
- All users (students/staff) must have a valid UTM ID to access the portal.
- Initial locker hardware setup and maintenance are assumed to be handled by UTM facilities.
- Size of each locker are enough to accommodate different size of parcels

6.1 System Boundaries

The system boundary diagram below illustrates the scope of the Cengal Parcel Point system. It shows the system (inside the boundary box) and the external entities that interact with it.

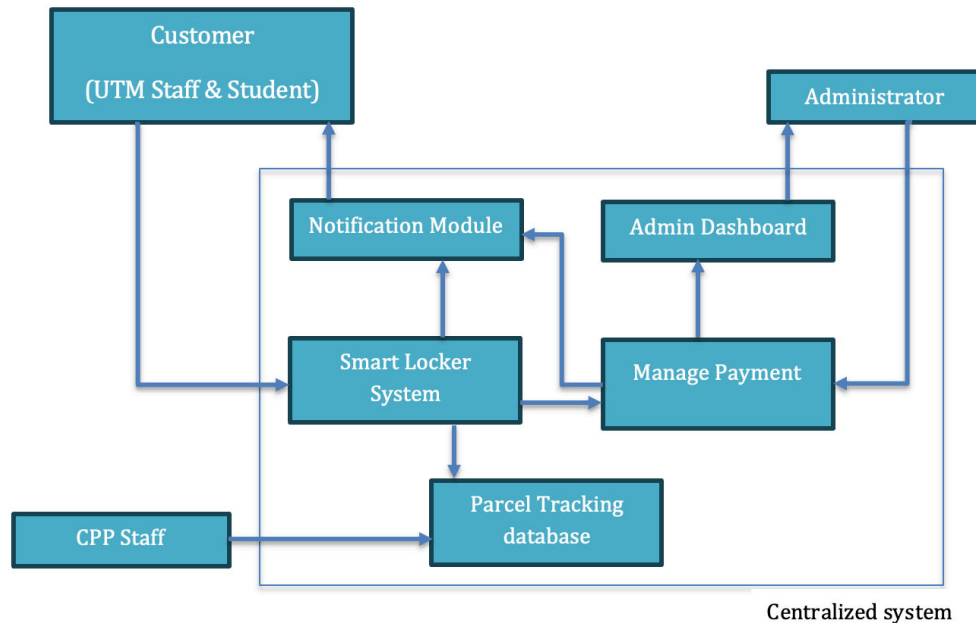


Figure 6.1 System boundary for the Cengal Parcel Point Database System

The system boundary defined for the Cengal Parcel Point (CPP) Parcel Service System clearly distinguishes the internal processes managed by the database and the external entities that interact with it, such as customers, staff, administrators, and external courier services. By setting these boundaries, the system ensures focused data flow control, preventing unnecessary complexity while maintaining clear accountability between different user roles. The inclusion of modules such as parcel logging, automated notifications, and locker management within the system scope ensures that all critical internal operations are digitized and centralized. Conversely, excluding features such as third-party courier integration and mobile app development in this phase reflects a practical limitation based on time, cost, and resource feasibility. This boundary justification aligns with the project's objectives to enhance efficiency, security, and accessibility without overextending system capabilities beyond current operational needs. The well-defined system boundary also provides a structured foundation for future scalability, allowing additional modules or integrations to be implemented seamlessly in later development stages.

6.2 Major User View

There are three main users in this system: Customers (UTM Staff and Student), CPP Staff, and Administrators. The table below illustrates the access privileges of each major user group to the key data entities within the Parcel Service System.

DATA	Access Type	Administrator	CPP Staff	Recipients (Customer)
Parcel	Maintain	X	X	
	Query	X	X	X
	Report	X	X	
Customer (Student @ UTM Staff)	Maintain	X		(View Own Only)
	Query	X	X	
	Report	X		
Staff/Admin	Maintain	X		
	Query	X		
	Report	X		
Smart Locker	Maintain	X		
	Query	X	X	X (Status Only)
	Report	X	X	
Notification	Maintain	X (Templates)	X (Send)	
	Query	X	X	X (Receive Only)
	Report	X		
Payment (Penalty)	Maintain	X	X (Record)	X (Pay Own Only)
	Query	X	X	X (View Own Only)
	Report	X	X	

Table 6.2 Major User View Table

Administrator: Has full access to maintain, query, and report on all data entities. They manage system configurations, user accounts, and oversee overall operations.

CPP Staff: Manages and queries Parcel data as part of daily operations, assists customers, monitors locker status, triggers notifications, and records penalty payments.

Customer (UTM Staff and Student): Has limited access to view and manage their own parcel and payment information, check locker status, and receive notifications.

7.0 Project Planning

Project planning is a crucial phase that outlines the overall strategy for developing the Parcel Service System at Cengal Parcel Point (CPP). It ensures that the project progresses systematically from planning to implementation within the given timeframe. This section defines the team structure, roles, and responsibilities, followed by the work breakdown structure (WBS) and Gantt chart to guide the project's workflow. Proper planning helps the team manage time, resources, and deliverables efficiently while maintaining alignment with the project objectives.

7.1 Human Resource

Team Structure & Responsibilities

Role	Key Responsibilities	Skills Required	Time Commitment
Project Manager (1)	- Plan timelines & milestones - Coordinate team tasks - Conduct weekly progress meeting	Leadership, Organization	4–6 hrs/week
System Analyst (1)	- Gather requirements - Analyze current process - Define data requirements and workflows	System analysis, Documentation	4–8 hrs/week
Backend Developer (2)	- Integrate database with APIs - Implement notification and QR code modules	Python/Java, API integration	10–15 hrs/week
Frontend Developer (1)	- Develop web interfaces for admin and users - Integrate UI with backend	HTML, CSS, JavaScript	10–15 hrs/week
UI/UX Designer (1)	- Design interfaces for admin and User UTM portals - Prototype locker system	Figma/Adobe XD	6–10 hrs/week
QA Tester (1)	- Test database transactions, functionality and system integration - Report bugs	- Software Testing - Selenium	4–8 hrs/week
CPP Staff (2)	- Provide operational feedback - Test system usability - Assist during implementation	Parcel management knowledge	6-8 hrs/week

Database Developer (1)	- Design database schema - Create ERD - Implement and maintain MySQL database - Ensure data accuracy and backup	MySQL, Data modeling	8–10 hrs/week
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Table 7.1: Team Structure & Responsibilities

The success of the Parcel Service System project relies heavily on effective teamwork and proper role allocation among team members. Each member is assigned specific responsibilities based on their skills and expertise to ensure smooth project execution. This section outlines the key roles, responsibilities, and required skills for each team member, ensuring that all phases of the project from planning and analysis to implementation and testing are completed efficiently and collaboratively.

7.2 Work Breakdown Structure (WBS)

Level 1: Parcel Service System Project

Level 2: Project Phases

1.0 Planning

- 1.1 Understand organizational background (CPP Centre)
- 1.2 Identify system problem and needs
- 1.3 Define project objectives and scope
- 1.4 Identify technical and database requirements
- 1.5 Prepare project plan, milestones, and resource allocation

2.0 Analysis

- 2.1 Conduct interview with CPP staff and students
- 2.2 Analyze existing manual process (AS-IS system)
- 2.3 Identify system and data requirements
- 2.4 Develop data dictionary and determine entities/attributes
- 2.5 Conduct feasibility study (technical, operational, and economic)

3.0 Design

- 3.1 Design system architecture (frontend, backend, database)
- 3.2 Create Entity Relationship Diagram (ERD) for the database
- 3.3 Design database schema (tables, keys, relationships)
- 3.4 Design UI/UX for admin dashboard and student portal
- 3.5 Create Logical DFD (TO-BE system)

4.0 Implementation

4.1 Database Development

- 4.1.1 Create database using MySQL
- 4.1.2 Implement tables (Parcel, User, Locker, Payment, etc.)
- 4.1.3 Set up relationships and constraints (Primary and Foreign Keys)
- 4.1.4 Insert initial data for testing

4.2 Backend Development

- 4.2.1 Develop APIs for database communication
- 4.2.2 Implement notification and parcel tracking modules
- 4.2.3 Integrate QR code generation

4.3 Frontend Development

- 4.3.1 Develop user interface for student and admin portals
- 4.3.2 Connect frontend with backend database

4.4 Smart Locker Integration

- 4.4.1 Integrate locker API with system database
- 4.4.2 Automate locker status updates upon parcel pickup

5.0 Testing

- 5.1 Perform Database Testing (CRUD operations and data integrity)
- 5.2 Perform System Integration Testing (frontend–backend–database)
- 5.3 Conduct User Acceptance Testing (UAT) with CPP staff and students
- 5.4 Debug and optimize database queries and system performance

6.0 Deployment and maintenance

- 6.1 Deploy system and database to UTM local server/cloud
- 6.2 Conduct training session for CPP staff
- 6.3 Perform database backup and maintenance plan
- 6.4 Collect feedback and perform continuous improvements

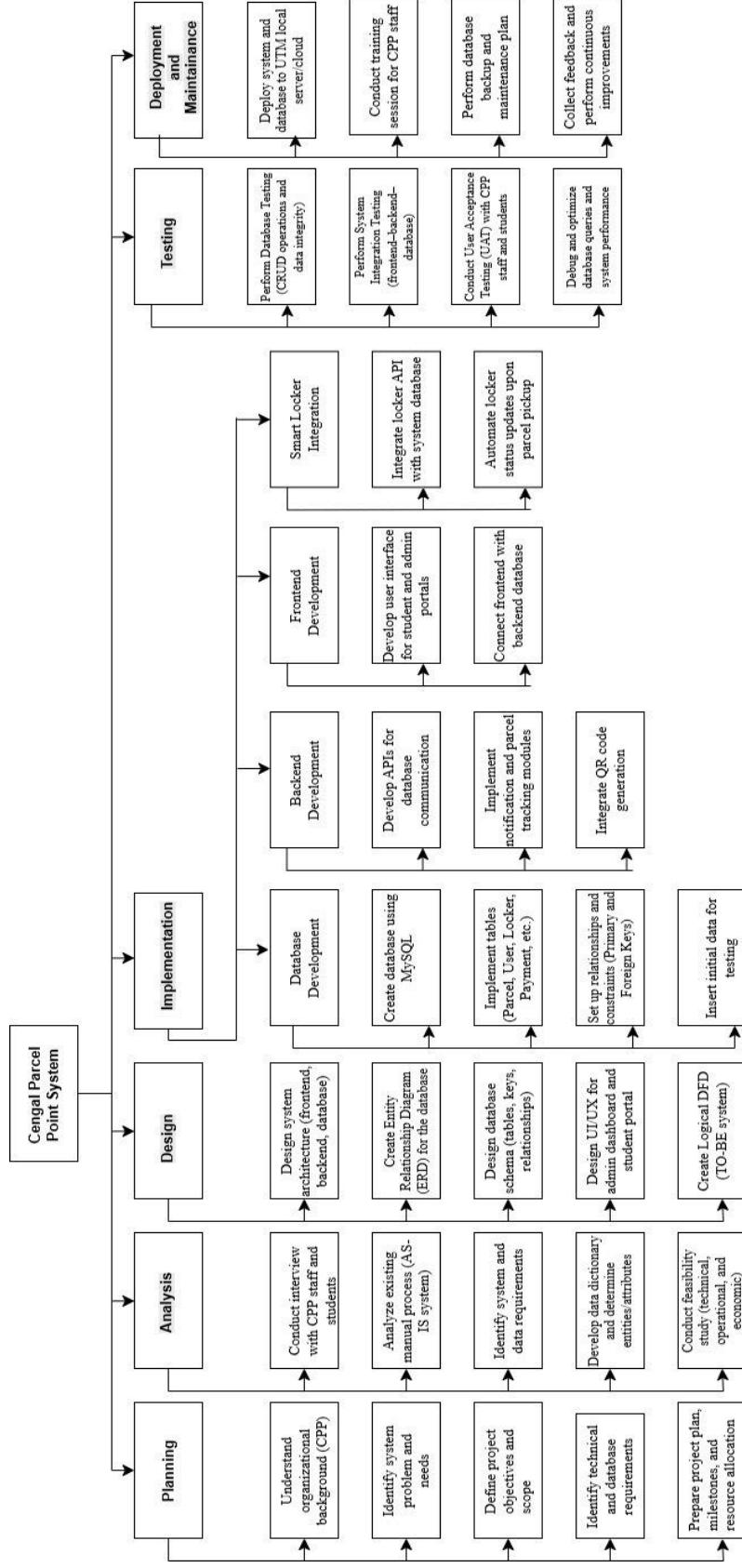


Chart 7.2: Work Breakdown Structure

https://app.diagrams.net/?src=about#GIFvFuwPR4nFMICTzn60_uV9bDqzSPi_kh#%7B%22pageId%22%3A%222yTOOgI3jGWBRLcH376G%22%7D

7.3 Gantt Chart

CENGAL PARCEL POINT SYSTEM

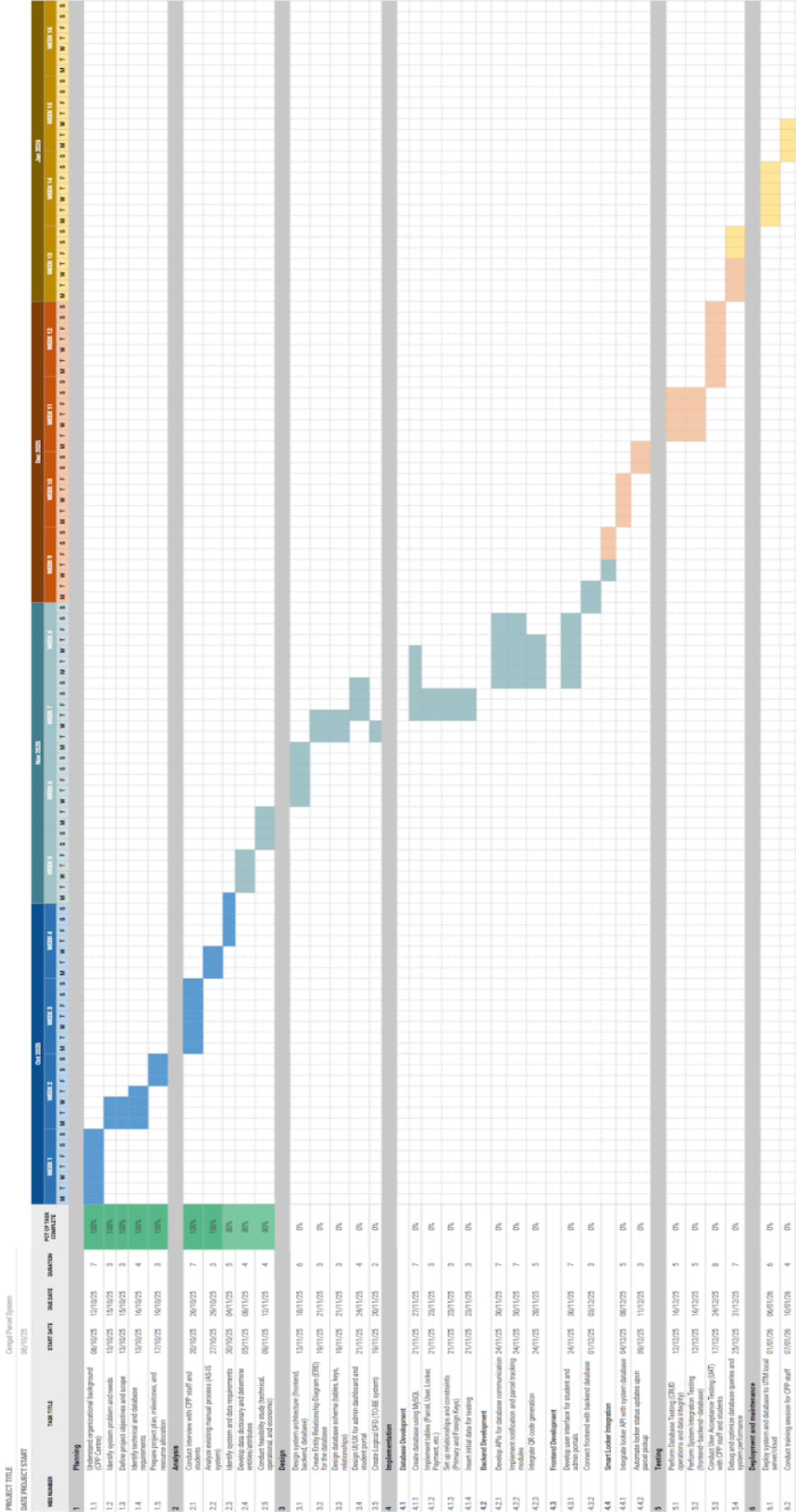


Figure 7.4.1: Gantt Chart for system development

<https://docs.google.com/spreadsheets/d/1wQWHscOx22JpGWJgPGPe2PiCBePJVJUrIg8UPRU-tmY/edit?usp=shari>

8.0 Requirement Analysis

8.1 Current business process

1. Current process overview:

The current parcel management at the university's Cengal Parcel Point (CPP) is handled manually. When couriers deliver parcels, the staff will record the parcel details on a computer using an Excel file or writing on a logbook physically. Furthermore, notification about parcels ready to be picked up is notified via WhatsApp or sometimes students need to walk in to check whether their parcels have arrived or not. The process is heavily dependent on human effort including written records and informal communication.

There is no centralized, automated and efficient system to track parcel status, student verification, or deliver records. The manual process results in inefficiencies, delays and a lack of accountability due to the increase of parcel volumes nowadays.

2. Detailed Workflow Steps (AS-IS)

i. Parcel Arrival:

- Courier services like Shopee Express which is mainly dropping parcels to Cengal Parcel Point (CPP) office. Sometimes also receiving parcels from different courier companies such as J&T, PosLaju.
- Parcels are handed over to the staff without being scanned into any formal and automated organized system.

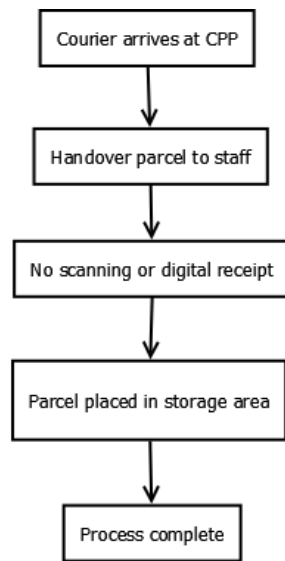


Figure 8.1.1: Flow Chart for parcel arrival process

ii. Manual Recording:

- Staff manually records parcel information such as recipient's name, contact number, tracking number, and time of arrival.
- This data is either written in a logbook or entered into the computer Excel sheet.

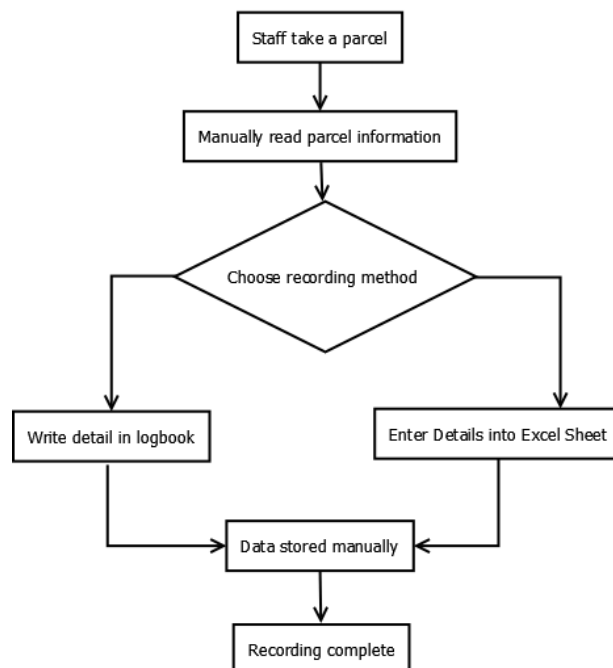


Figure 8.1.2: Flow Chart for Manual Recording Process

iii. Notification process:

- Recipients like students and university staff are normally notified through WhatsApp messages to ask them to come and pick up their parcels once ready.
- In a few cases, recipients will walk in to the office and check their parcel status if not receiving any notification from CPP.

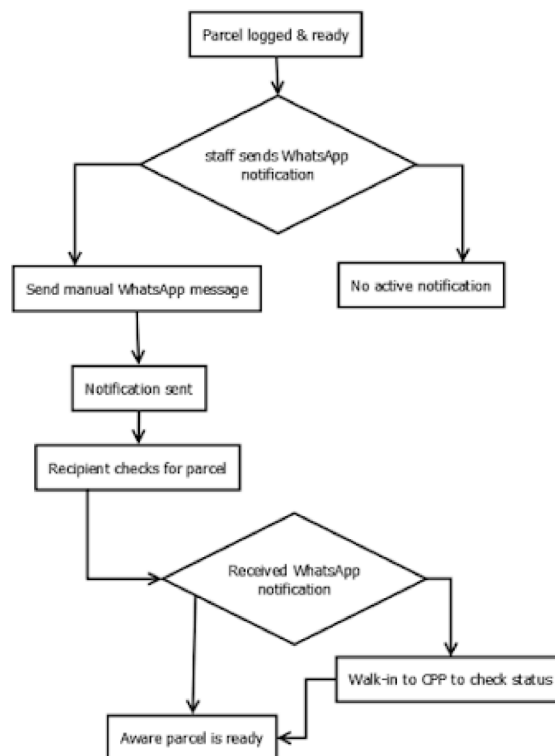


Figure 8.1.3: Flow Chart for notification process

iv. Parcel Collection by Recipients:

- Recipients need to come at office hours to collect their parcels.
- They need to provide a phone number, tracking number and name for staff to verify and check their matching parcel details.
- Once a parcel is found and verified, the staff will hand over the parcel.

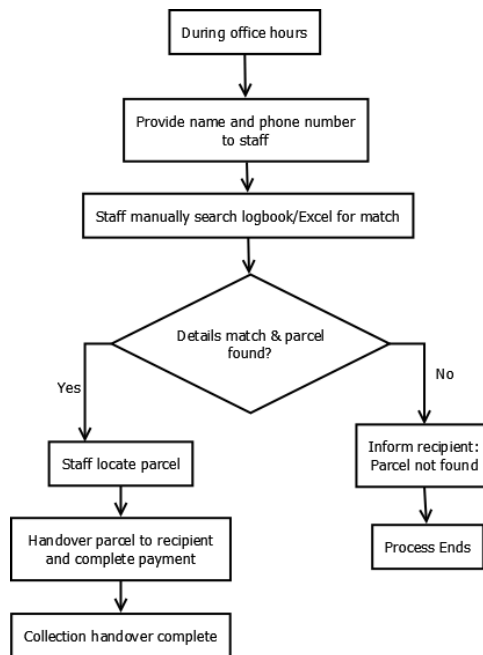


Figure 8.1.4: Flow Chart for parcel collection by recipients

v. Manual Update for Parcel Status:

- Once a parcel is collected, the staff ticks or crosses it out from the record list to indicate the parcel has been taken.
- There is no systematic log of collection times or confirmation from recipients which may cause error when recording.

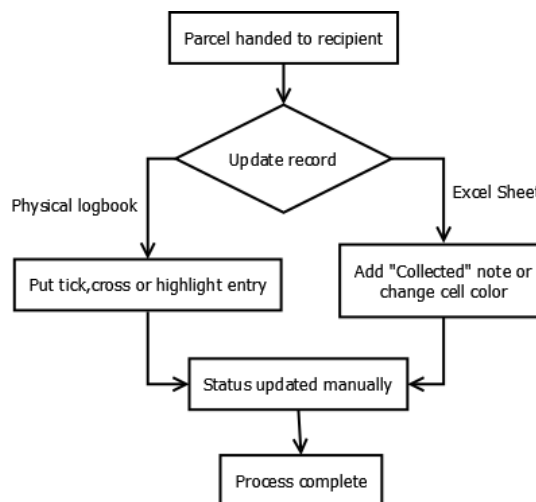


Figure 8.1.5: Flow Chart for manual update of parcel status process

3. Stakeholders Involved

- Courier Staff from Shopee Express, J&T, PosLaju and so on delivering parcels to the CPP office.
- Parcel Management Staff (CPP) who are responsible for receiving, recording, storing and handling over parcels.
- Students or staff from University Technology Malaysia as recipients of parcels.
- Administrative officers occasionally assist during busy periods or when problems occur.

4. Tools and Technologies Used

- Physical logbook or Excel spreadsheet to record parcel details.
- WhatsApp messages to inform recipients once the parcel is ready to be picked up.
- Phone number, tracking number and name used for informal identity verification.
- No barcode scanner, digital confirmation, or real-time parcel tracking system.

5. Problems Identified and Process Challenges

- High manual workload like staff handling many parcels daily without an automated and organized system.
- High chance of errors in manual recording.
- Recipients
- No system to track which parcels are pending/uncollected.
- Inconvenient and slow during peak parcel collection times.

8.2 General Requirement

1. **Performance Requirements** are non-existent in the current system, as its speed is entirely dependent on manual labor. The response time for logging a parcel or answering a recipient's status inquiry is slow, often taking several minutes as staff must manually search through logbooks or spreadsheets. This process creates significant delays and long queues, especially during peak hours when the volume of parcels is high.
2. **Security Measures** are minimal and fundamentally inadequate for protecting sensitive information. There is no user authentication for staff to access parcel records, and all data is stored in an unsecured physical logbook or an unprotected computer file. The verification of recipients during collection is also weak, relying only on a verbal confirmation of a name and phone number, which offers little protection against errors or misuse.
3. **Reliability and Backup** features are virtually non-existent, making the system highly vulnerable to failure. The entire operation depends on a single point of failure—the master logbook or spreadsheet—with no formal or automated backup procedures in place. In the event this record is lost or corrupted, there is no recovery plan, and all parcel data would be irrecoverably lost.
4. **Scalability and Limitations** prevent the system from handling increased demand effectively. The manual processes are linear and cannot scale; a higher parcel volume directly results in more work, longer delays, and a greater number of errors. This human bottleneck imposes a hard limit on the number of parcels that can be processed daily, making the system unsuitable for future growth.

8.3 Functional Requirement

These requirements describe the core functionalities provided by the manual system at Cengal Parcel Point (CPP). This system depends on human effort, handwritten on Excel logs and informal communication tools.

FR1 Parcel Logging

FR1.1 CPP staff manually receive parcels from couriers without using a tracking or scanning system.

FR1.2 The staff record parcel details (name, contact number, tracking number) in a logbook or Excel spreadsheet.

FR1.3 No data validation is performed, increasing the risk of duplicate or incorrect entries.

FR1.4 Parcel information is stored locally without central access or backup mechanisms.

FR1.5 Staff manually cross out or tick off parcels upon collection, without digital confirmation.

FR2 Notification Feature

FR2.1 Staff manually send WhatsApp messages to notify students or staff when parcels are ready.

FR2.2 Notifications are inconsistent and may be delayed or misdirected.

FR2.3 No structured or automatic re-notification is in place.

FR2.4 Some recipients receive no notification and must walk in to check manually.

FR3 Recipient Interaction Process

FR3.1 Students must visit the CPP office physically to check or collect parcels.

FR3.2 Recipients are asked to provide name, phone number, and tracking number for manual verification.

FR3.3 The status of a parcel is not updated in real-time or visible to students.

FR3.4 There is no portal or interface for students to track parcel status.

FR3.5 There is no system for late pickup reminders or overdue penalties.

FR4 Parcel Collection Process

FR4.1 Parcel collection is only available during CPP's working hours.

FR4.2 Staff manually search for parcels in storage areas based on physical records.

FR4.3 Parcel handover is verified verbally or using paper records.

FR4.4 No official time-stamping or digital confirmation is used during parcel pickup.

FR5 Storage and Monitoring

FR5.1 Storage is managed manually by placing parcels on shelves without digital tracking.

FR5.2 No real-time awareness of storage capacity or parcel status.

FR5.3 No automated alerts for uncollected or overdue parcels.

FR5.4 Reports are generated manually, if at all

8.4 Non-functional Requirement

These non-functional requirements reflect the operational limitations, user experience challenges, and security concerns of the current manual system.

NFR1 Performance of the system

NFR1.1 The system relies on staff to handle ~100 parcels per day manually, leading to delays during peak times.

NFR1.2 Notification delivery time is inconsistent and depends on staff availability.

NFR1.3 No real-time updates for parcel status; information is static and often outdated.

NFR1.4 Parcel collection and verification processes can take several minutes per user.

NFR2 Control Requirement

NFR2.1 No role-based access control — all staff may access or modify records.

NFR2.2 No audit trails or logs for actions taken on parcels..

NFR2.3 No central system for monitoring parcel flow or overdue pickups.

NFR2.4 No control mechanisms in place during emergencies (e.g., lost parcel, system failure).

NFR2.5 Reports and performance summaries are rarely generated and are not standardized.

NFR3 Security and Data Privacy

NFR3.1 User and parcel data is not encrypted — vulnerable to loss, misplacement, or unauthorized access.

NFR3.2 WhatsApp and Excel are not secure communication or data management tools.

NFR3.3 Identity verification is informal and error-prone.

NFR3.4 No data backup practices are in place for Excel files or logbooks.

NFR4 Usability and Accessibility

NFR4.1 Manual records are difficult to navigate and maintain under high volume.

NFR4.2 System is only accessible physically at CPP during office hours.

NFR4.3 No user interface or portal; all queries must be handled in person.

NFR4.4 Students often need to ask staff directly, causing long queues and inconvenience.

NFR5 Scalability and Maintainability

NFR5.1 The manual system does not scale — high parcel volume overwhelms storage and staff.

NFR5.2 System maintenance (e.g., updating Excel, correcting errors) is time-consuming and error-prone.

NFR5.3 There is no user or staff training documentation or system manual.

8.5 Logical DFD AS-IS system

Context diagram

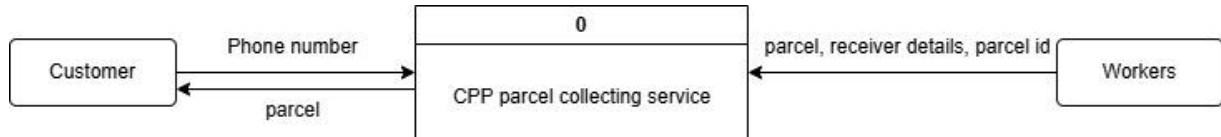


Figure 8.5.1: Context Diagram of the AS-IS system

Diagram 0

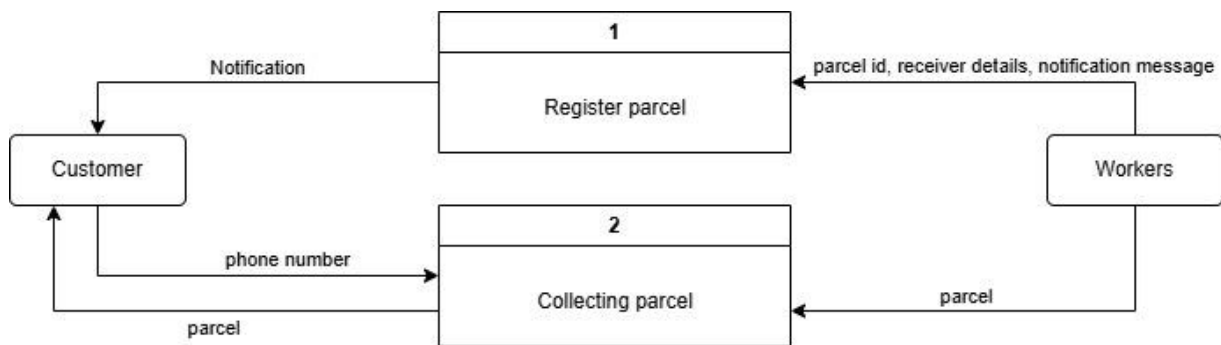


Figure 8.5.2: Diagram 0 of the AS-IS system

8.6 Business Rules (Current System)

The current parcel management process at Cengal Parcel Point (CPP) operates manually with limited automation. The following business rules define how staff, customers, and other entities interact under the existing workflow:

1. Parcel Handling Rules

- All incoming parcels must be recorded manually by CPP staff in a logbook or Excel sheet.
- Each parcel entry must include the recipient's name, phone number, courier name, and tracking number.
- No parcel may be released without recipient verification through name, phone number, or tracking ID.
- Staff are responsible for organizing and storing parcels in the designated storage area according to size or arrival date.

2. Notification Rules

- Notifications to recipients are sent manually through WhatsApp once a parcel is ready for collection.
- If the message is sent incorrectly, staff must resend the correct notification manually.
- No automated re-notification or system confirmation is in place for undelivered or unread messages.

3. Collection Rules

- Recipients must collect their parcels in person during CPP office hours only.
- Verification requires providing name, phone number, or tracking number to staff.
- Staff manually confirm collection by ticking or crossing the record in the logbook or Excel file.
- If a parcel remains uncollected beyond the allowed period, staff may impose a penalty fee.

4. Payment and Penalty Rules

- Penalty fees for late parcel collection are recorded manually by staff.
- Payment transactions are made in cash and documented separately from parcel logs.
- There is no automatic linkage between parcel status and payment record.

5. Record Maintenance Rules

- Only CPP staff are allowed to create, update, or delete parcel records.
- Corrections to recorded information (e.g., wrong contact or parcel ID) must be done manually.
- There is no centralized database or version control to track changes.
- Reports on parcel volume or uncollected parcels are prepared manually using recorded data.

8.7 Data Requirement

1. Types of Data Currently Stored (AS-IS)

- **Customer Information:** Name and contact details of parcel recipients.
- **Parcel Details:** tracking number, receiver name, receiver phone number and arrival date.
- **Notification Record:** receiver phone number, arrival date, parcel details
- **Payment:** collection fees record, penalty for overdue parcel record

2. How Data Is Used in Operations / Reports

Data Type	Usage in Operations / Reports
Customer	Verify student identity, contact for pickup notifications, generate user activity reports.
Parcel	Track parcel progress, assign lockers, update status, generate parcel volume and handle reports.
Smart Locker	Check availability for parcel assignment, update locker status (Occupied, Available, Out-of-Order), generate locker utilization reports.
Receipt / Payment	Verify and record penalty payments, generate revenue and outstanding fee reports.
Report	Summarize operational performance, including collection efficiency and financial summaries for management review.

3. User Roles Accessing the Data

User Role	Data Accessed	Purpose
Customer (UTM Staff & Student)	Their own parcel status, locker assignment (QR Code), and payment history.	Check parcel status, receive pickup codes, and view or pay penalties.
CPP Staff	Parcel, Customer, Smart Locker, and Payment data.	Process incoming parcels, assign lockers, update collection status, manage locker maintenance, and record penalty payments.

Administrator	All data (Customer, Parcel, Locker, Payment, Report), summarized Parcel and Payment data.	Full system oversight, generate reports, manage users, troubleshoot issues, and configure system settings ,monitor system performance, collection rates, and financial summaries.
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4. Gaps / Issues with Data Handling

- Risks and Potential for Manual Errors: Although parcel handling is automated, initial data entry (e.g., parcel details) could still cause errors if not validated.
- Lack of centralized system: The system relies on manual logs or spreadsheets, leads to fragmented data storage and makes information retrieval difficult.
- Delayed Data updated: Manual processes cause delays in updating parcel
- Inefficient Workflow: The manual process consumes additional time for verification and report generation.

5. Data Elements to be Stored in Database(TO-BE)

Entity	Attributes
Parcel	parcelID, senderName, receiverName, receiverID, receiverContact, weight, lockerID, arrivalDate, pickupCode, status
Locker	lockerID, location, lockerStatus, capacity, lockerType
Customer (Student / Staff)	customerID, name, email, phoneNumber, role (student/staff), assignedLocker, parcelID
Account / Payment	paymentID, parcelID, paymentMethod, paymentAmount, paymentDate, paymentStatus, receiptNo
Management (Admin / CPP Staff)	adminID, username, password, role, activityLog, lastLogin
Report	reportID, reportDate, reportType, totalParcels, pendingParcels, collectedParcels, performanceSummary

6. Relationship Among Data

Data Entities	Relationship	Description
Parcel and Smart Locker	A parcel is stored in a Smart Locker.	Each parcel is assigned to one available locker. One locker holds one parcel at a time to ensure proper tracking.
Parcel and Customer (UTM Staff and Student)	A parcel is addressed to a Customer.	Each parcel links to a student via UTM ID, ensuring accurate notification and delivery.
Parcel and Receipt / Payment	A parcel may generate a Payment.	Payments are recorded when penalties apply (e.g., overdue pickup), linking each payment to its corresponding parcel.
Report and Parcel & Payment	A report summarizes Parcel and Payment data.	Reports aggregate operational and financial data, such as parcel counts, collection rates, and revenue.

9.0 Transaction Requirements (Entry, Update/Delete, Queries) (Proposed System)

9.1 Data Entry

1. Enter the details of a new customer and their parcel such as customer name, UTM ID, phone number, address, shipping number and parcel size to create a new parcel delivery record.
2. Enter the payment details such as collection fee, late fees to create a payment record
3. Enter the finalized payment information such as total fee to generate a receipt record for the customer.
4. Enter the aggregated data to create a performance report record for administrative review.

9.2 Data Update/Delete

1. Update the parcel status and customer payment information upon parcel collection, including applying any penalty fees.
2. Update the payment status from 'Pending' to 'Completed' once a payment is successfully processed.
3. Update the smart locker status to 'Available' once a parcel has been collected from it.
4. Delete inactive customer data once collection of parcel has been done and all payments have been settled

9.3 Data Queries

1. Display parcel information, collection status, and payment amount
2. Display customer details and parcel collection history
3. Display administrative report information, such as collection rates and payment amount
4. Display smart locker availability status and location for assigning new parcels.

10.0 Benefit of the Proposed System

The benefits of the proposal system included :

1. Reduced Staff Workload
 - Automating parcel logging, notification, and collection tracking reduces manual administrative tasks and allows staff to focus on higher-level operations.
2. Improved Accuracy and reduce human error
 - QR code integration and digital records minimize the chances of misplaced parcels or wrong deliveries.
3. Enhanced Security and Accountability
 - Secure pickup methods, identity verification, and system logs provide better security and accountability, ensuring parcels are collected only by the intended recipients.
4. 24/7 Accessibility via Smart Lockers
 - Self-service lockers allow students and staff to collect their parcels anytime, increasing flexibility and reducing congestion during peak hours.
5. Efficient Communication
 - Automated notifications through email, SMS, or the UTM portal ensure timely and consistent updates, replacing error-prone manual messages.
6. Optimized Storage and Space Utilization
 - The admin dashboard with real-time data on parcel turnover and overdue pickups allows better decision-making and storage management.
7. Increased User Satisfaction
 - The improved speed, accessibility, and transparency of the system will lead to a better user experience for both students and staff.
8. Scalability for Future Growth
 - The system is designed to handle increasing parcel volumes without the need to proportionally increase human resources, making it sustainable in the long run.

11.0 Summary

The proposed Parcel Service System for Cengal Parcel Point (CPP) at Universiti Teknologi Malaysia aims to transform the current manual parcel management process into a fully automated and database-driven digital solution. Through the integration of a centralized MySQL database, the system enables efficient parcel logging, real-time tracking, and secure data management for both staff and students.

By replacing manual Excel records and WhatsApp notifications with a web-based platform, the system ensures higher accuracy, faster processing, and better communication between CPP staff and parcel recipients. The use of QR code verification and smart locker integration further enhances parcel security, reduces congestion at the collection counter, and provides 24/7 accessibility to users.

In addition, the project's implementation plan is outlined through a detailed Work Breakdown Structure (WBS) which covers every stage from planning and requirement analysis to database design, system development, testing, and deployment. This structured workflow ensures that each component, including the database system, functions seamlessly together to achieve the project objectives.

Overall, the Parcel Service System delivers a reliable, scalable, and user-friendly solution that improves efficiency, enhances service quality, and strengthens accountability at Cengal Parcel Point. It not only reduces staff workload and human error but also offers a modernized, convenient, and secure experience for all users within the university environment.